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LAB OBSERAVTION

15. Perform EDA on iris dataset: dimensions, summary, standard deviation, quantiles, grouping by Species, pivot table, categorical grouping with Sepal.Length categories.

Answer)

R Code:

```
# Load dataset
data("iris")
# Display dimensions
print("Dimensions:")
print(dim(iris))
# Display summary
print("Summary:")
print(summary(iris))
# Standard deviation
print("Standard Deviation:")
print(sapply(iris[1:4], sd))
# Quantiles
print("Quantiles:")
```

```
print(sapply(iris[1:4], quantile))
# Mean measurements grouped by Species
species_mean <- aggregate(
  . ~ Species,
  data = iris,
  FUN = mean
)
print("Mean Measurements by Species:")
print(species_mean)
# Pivot table: average Sepal.Length by Species
pivot_table <- aggregate(
  Sepal.Length ~ Species,
  data = iris,
  FUN = mean
)
print("Pivot Table:")
print(pivot_table)
# Categorize Sepal.Length
iris$Sepal_Category <- cut(
  iris$Sepal.Length,
  breaks = c(-Inf, 5, 6, Inf),
  labels = c("Short", "Medium", "Long")
)
print("Categorical Grouping:")
print(table(iris$Species, iris$Sepal_Category))
```

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/Programs/Experiment-15 Exploratory Data Analysis Using the iris Dataset.
R")
[1] "Dimensions:"
[1] 150   5
[1] "Summary:"
  Sepal.Length    Sepal.Width     Petal.Length    Petal.Width          Species
 Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100   setosa    :50
 1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300   versicolor:50
 Median :5.800   Median :3.000   Median :4.350   Median :1.300   virginica :50
 Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
 3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
 Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
[1] "Standard Deviation:"
Sepal.Length  Sepal.Width Petal.Length  Petal.Width
   0.8280661    0.4358663    1.7652982    0.7622377
[1] "Quantiles:"
     Sepal.Length Sepal.Width Petal.Length Petal.Width
0%            4.3         2.0         1.00         0.1
25%           5.1         2.8         1.60         0.3
50%           5.8         3.0         4.35         1.3
75%           6.4         3.3         5.10         1.8
100%          7.9         4.4         6.90         2.5
[1] "Mean Measurements by Species:"
     Species Sepal.Length Sepal.Width Petal.Length Petal.Width
1     setosa        5.006       3.428        1.462       0.246
2 versicolor        5.936       2.770        4.260       1.326
3  virginica        6.588       2.974        5.552       2.026
[1] "Pivot Table:"
     Species Sepal.Length
1     setosa        5.006
2 versicolor        5.936
3  virginica        6.588
[1] "Categorical Grouping:"

            Short Medium Long
  setosa       28     22    0
  versicolor    3     27   20
  virginica     1      8   41
>
```

16. Explore USArrests dataset: summary statistics, state with largest rape arrests, max & min murder rates, correlation among features, states above median assault arrests and bottom 25% for murder, visualization with histogram, density, scatterplots, bar graphs.

Answer)

R Code:

```
# Load dataset

data("USArrests")

print("Summary Statistics:")

print(summary(USArrests))

# State with the largest Rape value

max_rape_state <- rownames(USArrests)[which.max(USArrests$Rape)]

cat("State with Largest Rape Value:", max_rape_state, "\n")

# Maximum and minimum Murder rates

cat("Maximum Murder Rate:", max(USArrests$Murder), "\n")

cat("Minimum Murder Rate:", min(USArrests$Murder), "\n")
```

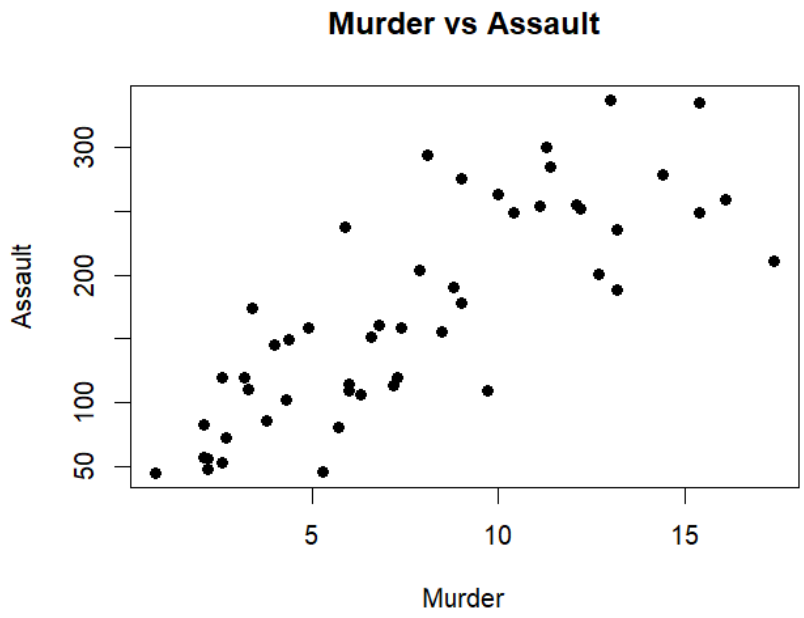
```
cat(
  "State with Maximum Murder Rate:",
  rownames(USArrests)[which.max(USArrests$Murder)],
  "\n"
)
cat(
  "State with Minimum Murder Rate:",
  rownames(USArrests)[which.min(USArrests$Murder)],
  "\n"
)
# Correlation matrix
print("Correlation Matrix:")
print(cor(USArrests))
# States above median Assault
median_assault <- median(USArrests$Assault)
above_median <- USArrests[
  USArrests$Assault > median_assault,
]
print("States Above Median Assault:")
print(above_median)
# States in the bottom 25% for Murder
q25 <- quantile(USArrests$Murder, 0.25)
bottom_murder <- USArrests[
  USArrests$Murder <= q25,
]
print("States in Bottom 25% for Murder:")
print(bottom_murder)
# Histogram
hist(
  USArrests$Murder,
```

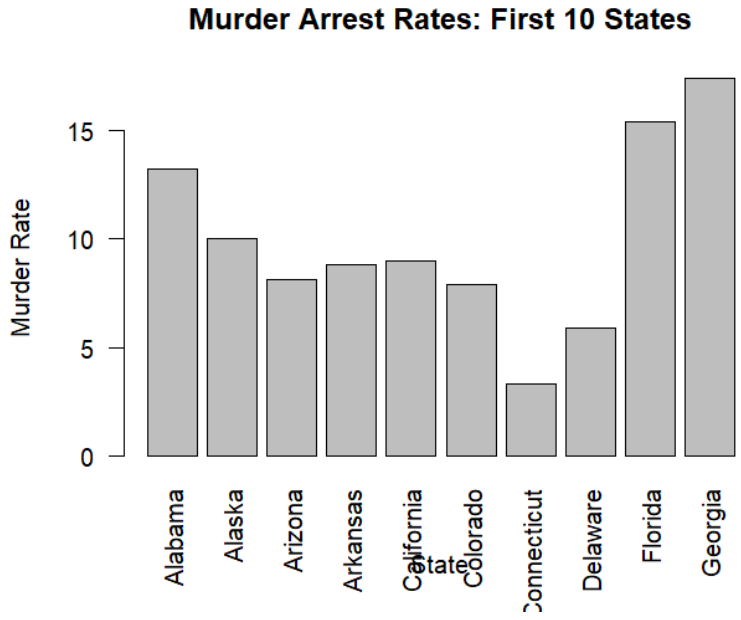
```
  main = "Histogram of Murder Arrest Rates",
  xlab = "Murder Arrest Rate"
)
# Density plot
plot(
  density(USArrests$Murder),
  main = "Density Plot of Murder Arrest Rates",
  xlab = "Murder Arrest Rate"
)
# Scatterplot
plot(
  USArrests$Murder,
  USArrests$Assault,
  main = "Murder vs Assault",
  xlab = "Murder",
  ylab = "Assault",
  pch = 19
)
# Bar graph of the first 10 states
barplot(
  USArrests$Murder[1:10],
  names.arg = rownames(USArrests)[1:10],
  main = "Murder Arrest Rates: First 10 States",
  xlab = "State",
  ylab = "Murder Rate",
  las = 2
)
```

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/Programs/Experiment-16 Explore the USArrests Dataset.R")
[1] "Summary Statistics:"
     Murder          Assault         UrbanPop          Rape
 Min.   : 0.800   Min.   : 45.0   Min.   :32.00   Min.   : 7.30
 1st Qu.: 4.075   1st Qu.:109.0   1st Qu.:54.50   1st Qu.:15.07
 Median : 7.250   Median :159.0   Median :66.00   Median :20.10
 Mean   : 7.788   Mean   :170.8   Mean   :65.54   Mean   :21.23
 3rd Qu.:11.250   3rd Qu.:249.0   3rd Qu.:77.75   3rd Qu.:26.18
 Max.   :17.400   Max.   :337.0   Max.   :91.00   Max.   :46.00
State with Largest Rape Value: Nevada
Maximum Murder Rate: 17.4
Minimum Murder Rate: 0.8
State with Maximum Murder Rate: Georgia
State with Minimum Murder Rate: North Dakota
[1] "Correlation Matrix:"
             Murder   Assault   UrbanPop      Rape
Murder   1.00000000 0.8018733 0.06957262 0.5635788
Assault  0.80187331 1.0000000 0.25887170 0.6652412
UrbanPop 0.06957262 0.2588717 1.00000000 0.4113412
Rape     0.56357883 0.6652412 0.41134124 1.0000000
[1] "States Above Median Assault:"
               Murder Assault UrbanPop Rape
Alabama          13.2     236       58 21.2
Alaska           10.0     263       48 44.5
Arizona           8.1     294       80 31.0
Arkansas          8.8     190       50 19.5
California        9.0     276       91 40.6
Colorado          7.9     204       78 38.7
Delaware          5.9     238       72 15.8
Florida          15.4     335       80 31.9
Georgia          17.4     211       60 25.8
Illinois         10.4     249       83 24.0
Louisiana        15.4     249       66 22.2
Maryland         11.3     300       67 27.8
Michigan         12.1     255       74 35.1
Mississippi      16.1     259       44 17.1
Missouri          9.0     178       70 28.2
Nevada           12.2     252       81 46.0
New Mexico       11.4     285       70 32.1
New York         11.1     254       86 26.1

North Carolina   13.0     337       45 16.1
Rhode Island      3.4     174       87  8.3
South Carolina   14.4     279       48 22.5
Tennessee        13.2     188       59 26.9
Texas            12.7     201       80 25.5
Wyoming           6.8     161       60 15.6
[1] "States in Bottom 25% for Murder:"
               Murder Assault UrbanPop Rape
Connecticut       3.3     110       77 11.1
Idaho             2.6     120       54 14.2
Iowa              2.2      56       57 11.3
Maine             2.1      83       51  7.8
Minnesota         2.7      72       66 14.9
New Hampshire     2.1      57       56  9.5
North Dakota      0.8      45       44  7.3
Rhode Island      3.4     174       87  8.3
South Dakota      3.8      86       45 12.8
Utah              3.2     120       80 22.9
Vermont           2.2      48       32 11.2
Washington        4.0     145       73 26.2
Wisconsin         2.6      53       66 10.8
>
```





17. Explore Titanic dataset: bar chart of survival vs class, modify plot by gender, histogram of Age.

Answer)

R Code:

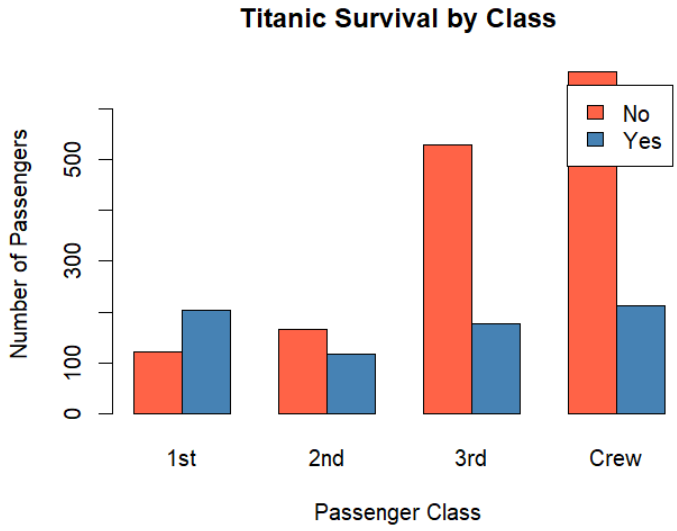
```
# Load the built-in Titanic dataset

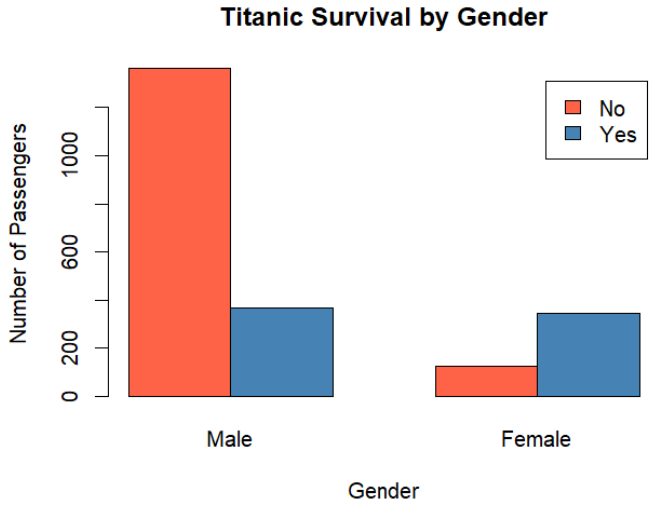
data("Titanic")
```

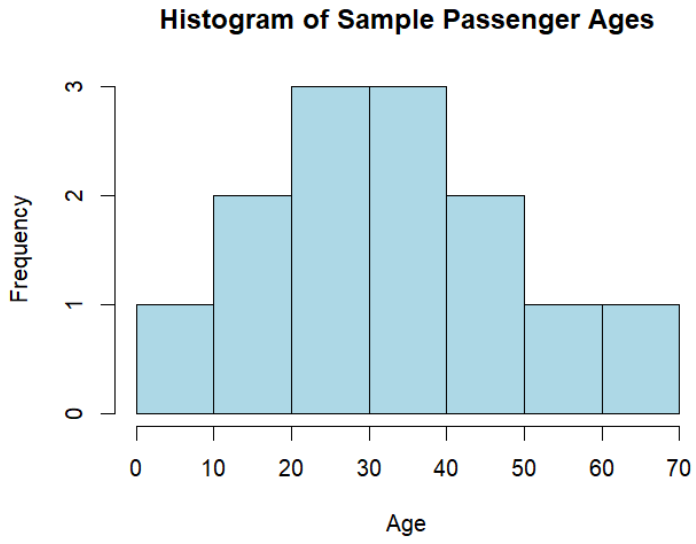
```
# Convert contingency table to a data frame
titanic_df <- as.data.frame(Titanic)
# Survival vs passenger class
survival_table <- xtabs(
  Freq ~ Class + Survived,
  data = titanic_df
)
barplot(
  t(survival_table),
  beside = TRUE,
  col = c("tomato", "steelblue"),
  legend.text = rownames(t(survival_table)),
  main = "Titanic Survival by Class",
  xlab = "Passenger Class",
  ylab = "Number of Passengers"
)
# Survival by gender
gender_table <- xtabs(
  Freq ~ Sex + Survived,
  data = titanic_df
)
barplot(
  t(gender_table),
  beside = TRUE,
  col = c("tomato", "steelblue"),
  legend.text = rownames(t(gender_table)),
  main = "Titanic Survival by Gender",
  xlab = "Gender",
  ylab = "Number of Passengers"
)
```

```
# The built-in Titanic table does not contain individual ages.
# Create a sample age vector for the histogram.
ages <- c(5, 12, 18, 22, 25, 28, 32, 35, 40, 45, 50, 60, 65)
hist(
  ages,
  breaks = 6,
  main = "Histogram of Sample Passenger Ages",
  xlab = "Age",
  col = "lightblue"
)
```

Output:







18. Create graphs in R: boxplot, histogram, bar plot, line chart, scatter plot.

Answer)

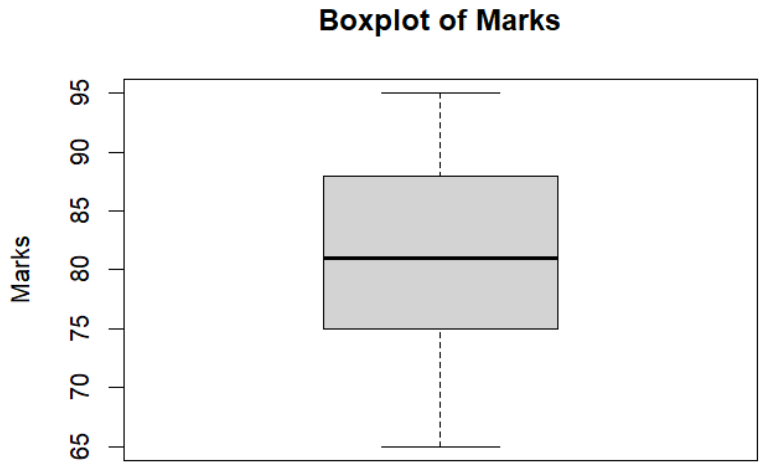
R Code:

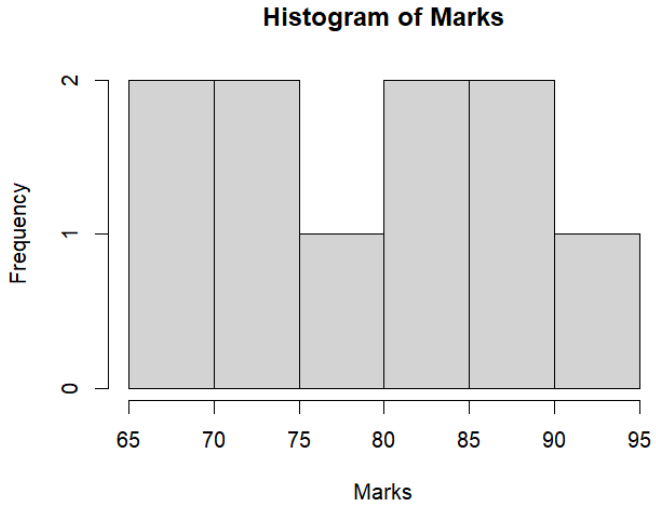
```
# Sample data
marks <- c(65, 70, 75, 80, 85, 90, 95, 75, 82, 88)
subjects <- c("Maths", "Science", "English", "Computer")
scores <- c(85, 90, 78, 95)
months <- 1:6
sales <- c(100, 120, 115, 140, 160, 180)
x <- c(1, 2, 3, 4, 5)
y <- c(10, 15, 13, 20, 25)
# Boxplot
boxplot(
  marks,
  main = "Boxplot of Marks",
  ylab = "Marks"
)
# Histogram
```

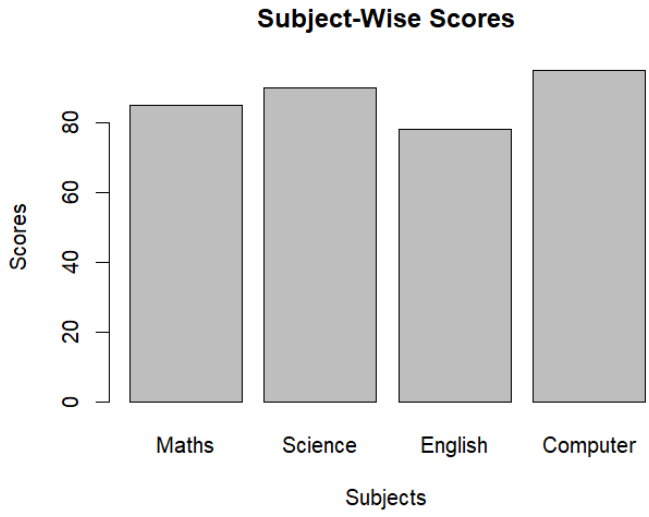
```
hist(
  marks,
  main = "Histogram of Marks",
  xlab = "Marks"
)
# Bar plot
barplot(
  scores,
  names.arg = subjects,
  main = "Subject-Wise Scores",
  xlab = "Subjects",
  ylab = "Scores"
)
# Line chart
plot(
  months,
  sales,
  type = "l",
  main = "Monthly Sales",
  xlab = "Month",
  ylab = "Sales"
)
# Scatterplot
plot(
  x,
  y,
  main = "Scatter Plot",
  xlab = "X Values",
  ylab = "Y Values",
  pch = 19
```

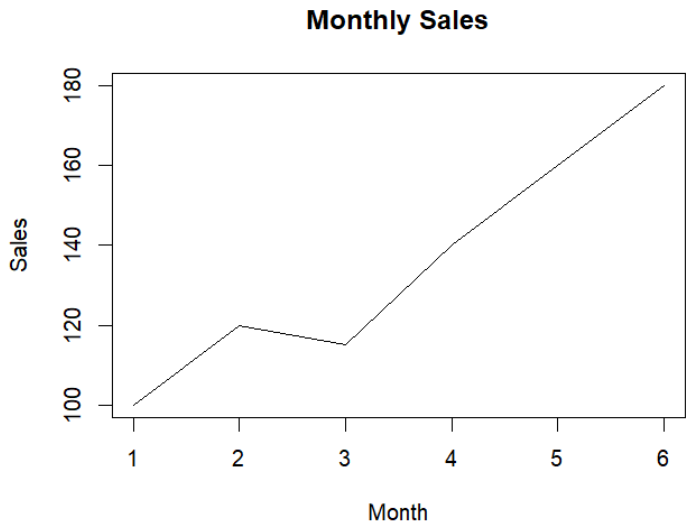
)

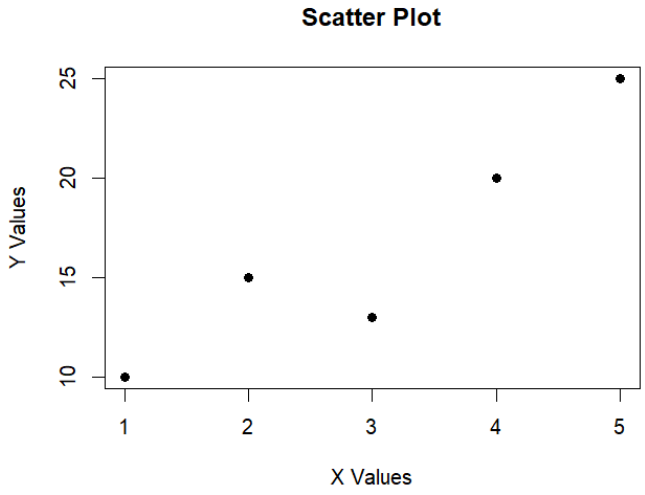
Output:


Boxplot of Marks
Marks
65
70
75
80
85
90
95


Histogram of Marks
Frequency
0
1
2
65
70
75
80
85
90
95
Marks


Subject-Wise Scores
Scores
0
20
40
60
80
Maths
Science
English
Computer
Subjects





19. Build a regression model on advertising dataset (Sales ~ Spend) and predict Sales.

Answer)

R Code:

```
# Create sample advertising data
advertising <- data.frame(
  Spend = c(10, 20, 30, 40, 50, 60, 70, 80),
  Sales = c(15, 25, 35, 42, 52, 61, 72, 80)
)
print("Advertising Dataset:")
```

```
print(advertising)

# Build linear regression model

model <- lm(Sales ~ Spend, data = advertising)

print("Regression Model Summary:")

print(summary(model))

# Predict sales for new advertising spend

new_data <- data.frame(Spend = 45)

prediction <- predict(model, newdata = new_data)

cat("Predicted Sales for Spend = 45:", prediction, "\n")
```

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/Programs/Eexperiment-19 Simple Linear Regression Using an
Advertising Dataset.R")
[1] "Advertising Dataset:"
  Spend Sales
1    10    15
2    20    25
3    30    35
4    40    42
5    50    52
6    60    61
7    70    72
8    80    80
[1] "Regression Model Summary:"

Call:
lm(formula = Sales ~ Spend, data = advertising)

Residuals:
    Min      1Q  Median      3Q     Max
-1.1191 -0.4464 -0.2500  0.5774  1.1429

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.07143    0.68367   8.881 0.000113 ***
Spend        0.92619    0.01354  68.411 6.56e-10 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8774 on 6 degrees of freedom
Multiple R-squared:  0.9987,    Adjusted R-squared:  0.9985
F-statistic:  4680 on 1 and 6 DF,  p-value: 6.563e-10

Predicted Sales for Spend = 45: 47.75
>
```

20. Create multiple regression model using ChickWeight dataset with “Time” and “Diet” as predictors; predict weight and compute model error.

Answer)

R Code:

```
# Load dataset
```

```
data("ChickWeight")
# Convert Diet to factor
ChickWeight$Diet <- factor(ChickWeight$Diet)
# Build multiple regression model
model <- lm(
  weight ~ Time + Diet,
  data = ChickWeight
)
print("Regression Model Summary:")
print(summary(model))
# Predict weight for all observations
predicted_weight <- predict(model, newdata = ChickWeight)
# Calculate mean squared error
actual_weight <- ChickWeight$weight
mse <- mean((actual_weight - predicted_weight)^2)
cat("Mean Squared Error:", mse, "\n")
# Predict weight for a new observation
new_chick <- data.frame(
  Time = 20,
  Diet = factor("2", levels = levels(ChickWeight$Diet))
)
new_prediction <- predict(model, newdata = new_chick)
cat("Predicted Weight:", new_prediction, "\n")
```

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/Programs/Experiement-20 Multiple Regression.R")
[1] "Regression Model Summary:"

Call:
lm(formula = weight ~ Time + Diet, data = ChickWeight)

Residuals:
     Min       1Q   Median       3Q      Max
-136.851  -17.151   -2.595   15.033  141.816

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  10.9244     3.3607   3.251  0.00122 **
Time          8.7505     0.2218  39.451  < 2e-16 ***
Diet2        16.1661     4.0858   3.957 8.56e-05 ***
Diet3        36.4994     4.0858   8.933  < 2e-16 ***
Diet4        30.2335     4.1075   7.361 6.39e-13 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 35.99 on 573 degrees of freedom
Multiple R-squared:  0.7453,    Adjusted R-squared:  0.7435
F-statistic: 419.2 on 4 and 573 DF,  p-value: < 2.2e-16

Mean Squared Error: 1284.319
Predicted Weight: 202.1003
>
```

21. Randomly split iris dataset into train/test (80/20), build logistic regression (Species ~ Petal.Length + Petal.Width), predict, and evaluate with confusion matrix.

Answer)

R Code:

```
# Install the package once if required

# install.packages("nnet")

library(nnet)

# Load dataset

data("iris")

# Set seed

set.seed(123)

# Randomly split dataset into 80% training and 20% testing

train_index <- sample(

  seq_len(nrow(iris)),

  size = floor(0.8 * nrow(iris))

)
```

```
train_data <- iris[train_index, ]
test_data <- iris[-train_index, ]
# Build multinomial logistic regression model
model <- multinom(
  Species ~ Petal.Length + Petal.Width,
  data = train_data,
  trace = FALSE
)
print("Model Summary:")
print(summary(model))
# Predict species
predictions <- predict(
  model,
  newdata = test_data
)
# Confusion matrix
confusion_matrix <- table(
  Actual = test_data$Species,
  Predicted = predictions
)
print("Confusion Matrix:")
print(confusion_matrix)
# Calculate accuracy
accuracy <- mean(predictions == test_data$Species)
cat("Accuracy:", round(accuracy * 100, 2), "%\n")
```

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/Programs/Experiment-21 Logistic Regression Using the iris
Dataset.R")
[1] "Model Summary:"
Call:
multinom(formula = Species ~ Petal.Length + Petal.Width, data = train_data,
    trace = FALSE)

Coefficients:
           (Intercept) Petal.Length Petal.Width
versicolor   -22.80429     6.592006    8.275971
virginica    -68.83031    12.810970   18.028532

Std. Errors:
           (Intercept) Petal.Length Petal.Width
versicolor    48.76232     41.02437    88.67880
virginica     51.06842     41.11363    88.76844

Residual Deviance: 17.18481
AIC: 29.18481
[1] "Confusion Matrix:"
            Predicted
Actual       setosa versicolor virginica
  setosa         10          0         0
  versicolor      0         14         1
  virginica       0          0         5
Accuracy: 96.67 %
>
```